SIMATS ENGINEERING

CO3- AT2 - Design Task - Medium

COURSE NAME: Advance Wireless Communication Systems /

5G / 6G Communication Systems

COURSE CODE: ECA56

COURSE OUTCOME COVERED

CO: 3 - *Design spectrum access strategies, waveform generation methods, and multiple access techniques for efficient wireless transmission systems.* (BL6)

Assessment Tool Weightage: 35%

Short description about assessment: Students design a system / model based on given requirements to assess creativity and application skills.

Dr.

QUESTIONS: 1 (1 X 100 = 100) **Total Marks: 100 Duration: 3hrs**

1. Design a system architecture for Multi-access Edge Computing to reduce latency in smart traffic applications. (BL6)
2. Develop a cloud-edge communication model for wireless healthcare monitoring. (BL6)
3. Design a functional architecture for real-time edge data processing in industrial automation. (BL6)
4. Propose a cloud-based wireless framework for smart surveillance systems. (BL6)
5. Design an edge AI workflow for intelligent wireless data filtering. (BL6)
6. Develop a task offloading strategy between edge and cloud in 5G applications. (BL6)
7. Design a low-latency communication architecture for edge-assisted smart city services. (BL6)
8. Propose a scalable edge-cloud framework for wireless sensor networks. (BL6)
9. Design a Cloud Radio Access Network model for dense urban deployment. (BL6)
10. Develop a functional split model between centralized and distributed radio units. (BL6)
11. Design a virtualized wireless access framework for next-generation communication. (BL6)
12. Propose a data flow model for cloud-based radio resource allocation. (BL6)
13. Design a centralized traffic control model using Cloud RAN principles. (BL6)
14. Develop a network management architecture for cloud-assisted wireless systems. (BL6)
15. Design a wireless service framework integrating virtual network functions.

(BL6)
16. Propose an optimized fronthaul communication model. (BL6)
17. Design a satellite-assisted communication model using Low Earth Orbit satellite for remote wireless coverage. (BL6)
18. Develop a comparative architecture for LEO, MEO, and GEO backhaul systems. (BL6)
19. Design a spectrum sharing model between LTE and 5G users. (BL6)
20. Propose a frequency allocation strategy for mixed wireless services. (BL6)
21. Design a dynamic spectrum access framework for urban wireless systems. (BL6)
22. Develop a communication architecture combining terrestrial and satellite networks. (BL6)
23. Design a spectrum efficiency improvement model for dense communication zones. (BL6)
24. Propose a channel reuse plan for multi-user wireless communication. (BL6)
25. Design an NB-IoT architecture for smart utility monitoring. (BL6)
26. Develop an LTE-M communication workflow for wearable monitoring systems. (BL6)
27. Design a low-power communication model for battery-operated IoT devices. (BL6)
28. Propose a device communication plan for long-life sensor networks. (BL6)
29. Design a 5G-enabled IoT framework for campus automation. (BL6)
30. Develop an energy-efficient wireless communication strategy for sensor nodes. (BL6)
31. Design an IoT communication flow for periodic environmental sensing. (BL6)
32. Propose a scalable wireless architecture for smart connected devices. (BL6)
33. Design a smart city communication framework for traffic monitoring and surveillance. (BL6)
34. Develop an energy monitoring communication model for public infrastructure. (BL6)
35. Design a smart factory information flow for predictive maintenance. (BL6)
36. Propose a robotic automation communication framework for industrial control. (BL6)
37. Design a remote patient monitoring communication architecture. (BL6)
38. Develop a wireless communication workflow for AI-assisted healthcare diagnosis. (BL6)
39. Design a smart irrigation communication model for crop monitoring. (BL6)
40. Propose a livestock tracking communication architecture using wireless sensors. (BL6)

Code of Conduct:

*I **Sudharshshini R (192512072)** certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Question 29: 5G-Enabled IoT Framework for Campus Automation

1. Understanding of Design

The proposed system is a 5G-enabled Internet of Things (IoT) framework designed to automate major campus services. The architecture connects distributed sensors, smart devices, cameras, controllers, and actuators through a 5G wireless network and an IoT edge gateway. The system is intended to monitor and control lighting, energy consumption, environmental conditions, water usage, security, classroom occupancy, and other campus facilities.

The main design objectives are reliable wireless connectivity, low communication latency, scalable device connectivity, efficient energy usage, centralized monitoring, and fast response to campus events. 5G provides the wireless connectivity required for a large number of IoT devices, while edge processing allows time-sensitive decisions to be made close to the devices. Cloud services can be used for long-term storage, analytics, dashboards, and campus-wide management.

The main constraints include a large number of heterogeneous IoT devices, wireless interference, battery limitations for sensor nodes, network congestion, cybersecurity risks, and the need to maintain reliable operation across classrooms, laboratories, hostels, administrative buildings, parking areas, and outdoor spaces.

2. Proposed 5G-Enabled IoT Architecture

The functional architecture is:

IoT Sensors / Smart Devices → 5G Access Network → IoT Gateway / Edge Node → Real-Time Analytics → Actuators / Local Control → Cloud Platform → Campus Management Dashboard

At the device layer, sensors collect information such as temperature, occupancy, light level, energy consumption, water level, air quality, and equipment status. Smart cameras and access-control devices provide security-related information. The 5G network connects these devices to nearby edge gateways. The edge layer filters and analyses data, identifies events, and sends commands to actuators. The cloud stores historical information and supports large-scale analytics and centralized campus management.

3. Functional Blocks

IoT Sensor and Device Layer: Contains environmental, occupancy, energy, water, security, and equipment sensors together with smart meters, cameras, and actuators.

5G Connectivity Layer: Provides wireless communication for campus IoT devices and supports scalable connectivity, low latency, and reliable data transfer.

IoT Gateway / Edge Layer: Aggregates device data, performs preprocessing and filtering, and provides local computing close to campus facilities.

Real-Time Processing Layer: Analyses sensor values, detects abnormal conditions, and generates immediate control decisions.

Automation and Actuation Layer: Controls lights, HVAC equipment, pumps, alarms, access systems, and other campus equipment.

Cloud and Application Layer: Provides storage, dashboards, historical analytics, reports, device management, and campus-wide monitoring.

Security and Network Management: Provides authentication, encryption, access control, device identity, monitoring, and network resource management.

4. Campus Automation Applications

Smart Lighting: Occupancy and light sensors automatically switch or dim lights according to room usage and daylight conditions.

Smart Energy Monitoring: Smart meters measure electricity usage and help identify high-consumption buildings and equipment.

Smart Classroom Automation: Occupancy and environmental sensors can control lighting, ventilation, and other classroom services.

Smart Security: Connected cameras, access-control devices, and intrusion sensors provide continuous monitoring and alerts.

Smart Water Management: Water-level and flow sensors can identify abnormal consumption and support automated pump control.

Environmental Monitoring: Temperature, humidity, and air-quality sensors provide information for healthy and comfortable campus spaces.

Smart Parking: IoT sensors can detect parking occupancy and provide availability information to campus users.

5. Evaluation of Design

The proposed framework has several important strengths. First, 5G provides a scalable wireless communication layer for a large number of connected campus devices. Second, edge processing reduces the need to send every raw sensor measurement to a remote cloud, improving response time for automation tasks. Third, the cloud provides centralized visibility across different campus buildings and services. The modular architecture also makes it possible to add new IoT applications without redesigning the entire system.

The design can improve energy efficiency by using real-time occupancy and environmental information to control campus equipment. It can also reduce operational effort by automating routine monitoring and generating alerts only when abnormal conditions occur.

However, the architecture has limitations. A large number of connected devices increases network-management complexity. Wireless interference and congestion can affect communication reliability. IoT devices may also have limited processing power, memory, and battery capacity. Security is a major concern because compromised devices could affect campus services. Edge gateways can also become failure points if redundancy is not provided.

6. Design Justification

5G is selected as the wireless connectivity technology because the campus environment may contain a large and diverse population of connected devices. The framework requires reliable connectivity, scalable device support, and low-latency communication for automation services. Edge computing is included because services such as access control, security alerts, lighting control, and equipment protection may require rapid local decisions.

The cloud is retained for functions that do not require immediate response. It provides centralized storage, historical analysis, dashboard services, reporting, and coordination between different campus facilities. Thus, the proposed architecture combines 5G connectivity, edge intelligence, and cloud services to balance latency, scalability, bandwidth, and centralized management.

7. Data Flow and Communication

A campus IoT sensor first measures a physical parameter such as occupancy, temperature, energy consumption, or water level. The measurement is transmitted over the 5G network to an IoT gateway or edge node. The edge node preprocesses the data and compares it with configured conditions. If an event requires immediate action, the edge node sends a command to the relevant actuator. A summary or event record is then forwarded to the cloud for storage and monitoring.

Simplified data flow:

Sensor → 5G Network → Edge Gateway → Filtering / Analysis → Decision → Actuator → Cloud Synchronization → Dashboard

8. MATLAB Simulation

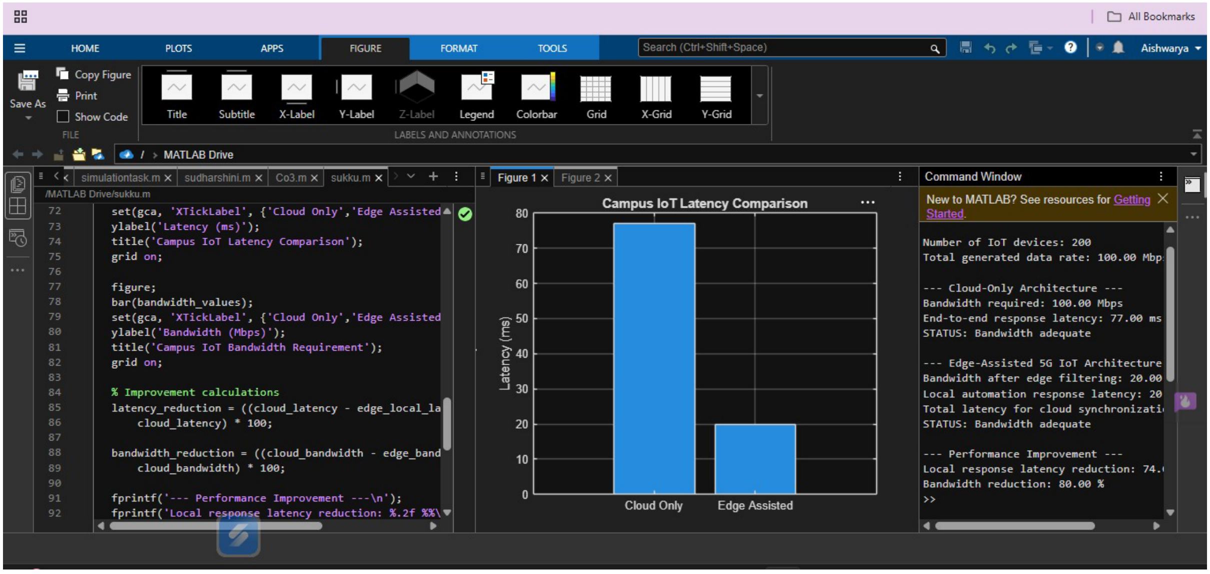
The MATLAB simulation evaluates the effect of edge-assisted 5G IoT processing on network traffic and response latency. The model considers 200 campus IoT devices, each generating an average of 0.5 Mbps. The total generated traffic is compared with a 300 Mbps network capacity. In the edge-assisted architecture, 80% of redundant sensor data is filtered locally. The simulation also compares cloud-only and edge-assisted response latency.

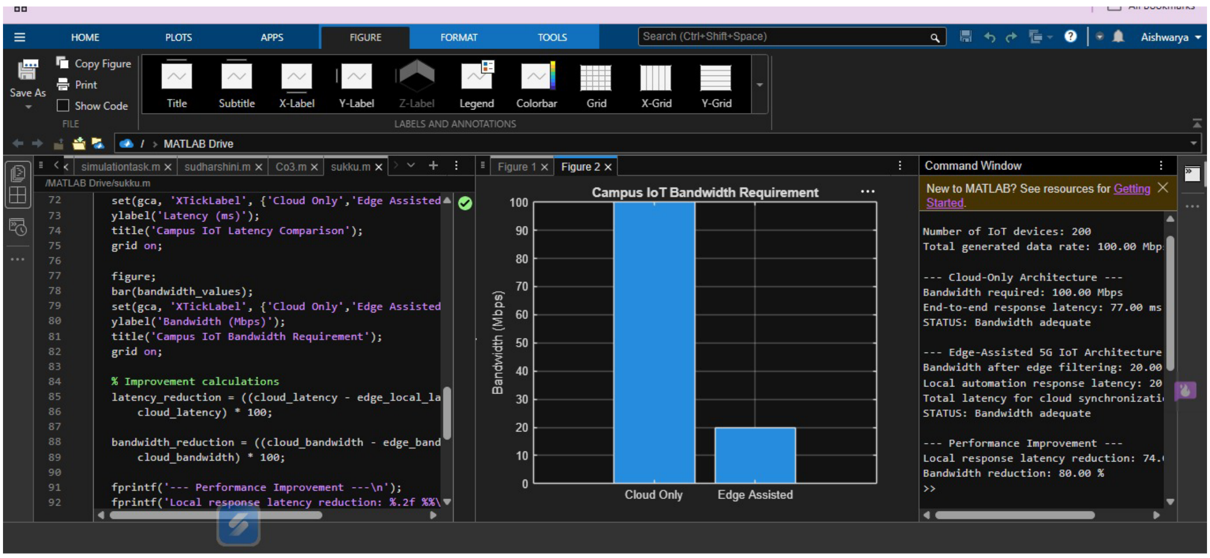
```
clc;
clear;
close all;

% 5G-Enabled IoT Framework for Campus Automation
% Cloud-only vs Edge-assisted IoT processing

% Campus IoT parameters
num_devices = 200;           % Number of IoT devices
data_rate_per_device = 0.5;   % Mbps per device
network_capacity = 300;       % Mbps

% Latency parameters in milliseconds
t_sensor = 2;                % Sensor acquisition delay
t_5g_cloud = 25;             % 5G + backhaul delay to cloud
t_cloud_processing = 50;      % Cloud analytics/processing delay

t_5g_edge = 3;               % 5G delay to nearby edge node
t_edge_processing = 15;       % Edge processing delay

% Edge filtering assumption
filtering_efficiency = 0.80;   % 80% redundant data removed

% Total generated traffic
total_data_rate = num_devices * data_rate_per_device;

% ---------------- Cloud-only architecture ----------------
cloud_bandwidth = total_data_rate;
cloud_latency = t_sensor + t_5g_cloud + t_cloud_processing;

% ---------------- Edge-assisted architecture -------------
edge_bandwidth = total_data_rate * (1 - filtering_efficiency);
edge_local_latency = t_sensor + t_5g_edge + t_edge_processing;

% Latency if the event is synchronized with cloud
edge_cloud_latency = edge_local_latency + ...
                     t_5g_cloud + t_cloud_processing;

fprintf('--- 5G IoT Campus Automation Simulation ---\n\n');
```

```
fprintf('Number of IoT devices: %d\n', num_devices);
fprintf('Total generated data rate: %.2f Mbps\n\n', total_data_rate);

fprintf('--- Cloud-Only Architecture ---\n');
fprintf('Bandwidth required: %.2f Mbps\n', cloud_bandwidth);
fprintf('End-to-end response latency: %.2f ms\n', cloud_latency);

if cloud_bandwidth > network_capacity
   fprintf('STATUS: CONGESTION - exceeded by %.2f Mbps\n\n', ...
      cloud_bandwidth - network_capacity);
else
   fprintf('STATUS: Bandwidth adequate\n\n');
end

fprintf('--- Edge-Assisted 5G IoT Architecture ---\n');
fprintf('Bandwidth after edge filtering: %.2f Mbps\n', edge_bandwidth);
fprintf('Local automation response latency: %.2f ms\n', edge_local_latency);
fprintf('Total latency for cloud synchronization: %.2f ms\n', ...
   edge_cloud_latency);

if edge_bandwidth > network_capacity
   fprintf('STATUS: CONGESTION\n\n');
else
   fprintf('STATUS: Bandwidth adequate\n\n');
end

% ---------------- Comparison graphs ----------------
latency_values = [cloud_latency edge_local_latency];
bandwidth_values = [cloud_bandwidth edge_bandwidth];

figure;
bar(latency_values);
set(gca, 'XTickLabel', {'Cloud Only','Edge Assisted'});
ylabel('Latency (ms)');
title('Campus IoT Latency Comparison');
grid on;

figure;
bar(bandwidth_values);
set(gca, 'XTickLabel', {'Cloud Only','Edge Assisted'});
ylabel('Bandwidth (Mbps)');
title('Campus IoT Bandwidth Requirement');
grid on;
```

```
% Improvement calculations
latency_reduction = ((cloud_latency - edge_local_latency) / ...
                    cloud_latency) * 100;

bandwidth_reduction = ((cloud_bandwidth - edge_bandwidth) / ...
                      cloud_bandwidth) * 100;

fprintf('--- Performance Improvement ---\n');
fprintf('Local response latency reduction: %.2f %%\n', latency_reduction);
fprintf('Bandwidth reduction: %.2f %%\n', bandwidth_reduction);
```

output:





9. Expected Simulation Interpretation:

For 200 IoT devices generating 0.5 Mbps each, the total traffic generated by the campus devices is 100 Mbps. In the cloud-only model, the complete 100 Mbps is transported toward the cloud. With 80% local filtering at the edge, only 20 Mbps is forwarded beyond the edge processing layer. This demonstrates a major reduction in the amount of data that must be transported for centralized processing.

Using the selected latency values, the cloud-only response latency is 77 ms, while the local edge response latency is 20 ms. Therefore, the edge-assisted architecture provides a substantially faster response for time-sensitive campus automation events. The reduction in forwarded data also leaves additional network capacity for other campus applications.

10. Suggestions for Improvement

• Deploy redundant edge gateways so that campus services continue operating if one gateway fails.

• Use adaptive data filtering based on occupancy, event severity, and network conditions.

• Use network slicing or prioritized resources for critical security and automation traffic.

• Implement strong device authentication, encryption, access control, and secure software updates.

• Use battery-aware communication strategies for low-power IoT sensors.

• Apply AI-based anomaly detection at the edge for energy, security, and equipment monitoring.

• Provide local buffering so important sensor events are retained during temporary network interruptions.

• Use centralized device management to monitor firmware, connectivity, faults, and security status.

11. Conclusion

The proposed 5G-enabled IoT framework provides a scalable architecture for campus automation by integrating IoT devices, 5G wireless connectivity, edge processing, actuators, and cloud services. The architecture enables real-time monitoring and control of lighting, energy, security, water, environmental conditions, classrooms, and parking. Edge processing reduces response latency and unnecessary data transmission, while the cloud provides centralized storage and campus-wide analytics. The MATLAB simulation demonstrates the benefits of local filtering in reducing bandwidth requirements and improving response time. With redundancy, secure communication, adaptive resource allocation, and intelligent analytics, the framework can support efficient and reliable smart campus services.

RUBRICS FOR CALCULATION:

Design Task					
Criteria	**Weightage**	**Level 4 (Exemplary)**	**Level 3 (Proficient)**	**Level 2 (Developing)**	**Level 1 (Beginning)**
Understanding of Design	20%	Demonstrates deep understanding of design objectives, constraints, and context	Shows good understanding with minor gaps	Partial understanding; misses key aspects	Lacks understanding of the design context
Evaluation of Design	25%	Provides in-depth critique identifying strengths, weaknesses, and design gaps	Identifies key issues with reasonable analysis	Limited critique; mostly descriptive feedback	No meaningful critique provided
Justification	20%	Supports critique with strong reasoning, evidence, and relevant principles	Provides justification with some supporting evidence	Weak or unclear justification	No justification for feedback
Suggestions for Improvement	20%	Offers innovative, practical, and well-justified improvement suggestions	Provides useful suggestions with minor gaps	Suggestions are limited or partially relevant	No constructive suggestions provided
Communication	15%	Communicates feedback clearly, professionally, and engages actively in discussion	Communicates well with minor issues	Communication lacks clarity or engagement	Poor communication and minimal participation